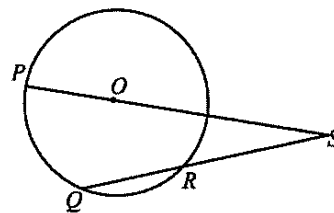


7. A number when divided by 899 gives a remainder 63. The remainder when this number is divided by 29 is
 a) 6 b) 7 c) 8 d) 5
8. A train leaves a station 1 hour before the scheduled time. The driver decreases the speed by 4 km/h. At the next station 120km away, the train reached on scheduled time. The original speed of the train is (in km/h)
 a) 24 b) 36 c) 18 d) 22
9. $ABCD$ is a square. From the diagonal BD , a length BX is cut off equal to BA . From X , a straight line XY is drawn perpendicular to BD to meet AD at Y . Then $AB + AY =$
 a) $\sqrt{2}BD$ b) $\frac{BD}{\sqrt{2}}$ c) $\sqrt{3}BD$ d) BD
10. The number of natural number pairs (x, y) in which $x > y$ and $\frac{5}{x} + \frac{6}{y} = 1$ is
 a) 1 b) 2 c) 3 d) 4
11. AB and AC are tangents at B and C to a circle. D is the mid point of the minor arc BC with respect to the triangle ABC , D is the
 a) orthocenter b) circumcentre c) incentre d) centroid
12. Two circles of radii in the ratio 1:2 touch each other externally. The centre of the small circle is C and that of the big is D . The point of contact is A . PAQ is a straight line where P is on the smaller circle and Q on the on the bigger circle (PAQ doesnot pass through C). The angle between the tangent at Q to the bigger circle and the diameter (produced if necessary) of the smaller circle is
 a) 60° b) 75° c) 80° d) none of these
13. The number of real solutions of the equation $\frac{|x-3| - |x+1|}{2|x+1|} = 1$ is
 a) 0 b) 1 c) 2 d) 3
14. The number of real x which satisfies the equal by $\frac{8^x + 27^x}{12^x + 18^x} = \frac{7}{6}$ is
 a) 2 b) 3 c) 4 d) 0
15. P, Q, R are there points on a circle of centre O . If $RS =$ radius of the circle and $\angle PSQ = 12^\circ$, then $\angle POQ =$
 a) 36° b) 42° c) 48° d) 54°



PART – B

Note:

- Write the correct answer in the space provided in the response sheet.
- For each correct response you get 1 mark; for each incorrect response you lose $\frac{1}{4}$ mark.

16. d is an integer greater than 1. When the numbers 1059, 1417 and 2312 are divided by ' d ', the remainders in each case is the same (r). The value of $(d-r)$ is _____.
17. The number of integers ' n ' which satisfy the inequality $(n^2 - 2)(n^2 - 20) < 0$ is _____.
18. The value of $\sqrt[3]{1+\sqrt{2}} \cdot \sqrt[6]{3-2\sqrt{2}}$ is _____.
19. When $a = 5$ and $b = 403$, the value of $\left\{ \frac{1}{a^2} + \frac{1}{b^2} + \frac{2}{a+b} \left(\frac{1}{a} + \frac{1}{b} \right) \right\} \left(\frac{(a+b)^2}{ab} \right)^{-1}$ is _____.
20. ABC is an isosceles triangle of base 30 cm. The altitude to the base is 20 cm. The length of the other altitude (in cm) is _____.
21. In a right triangle a semi-circle is inscribed so that its diameter lies on the hypotenuse and its centre divides the hypotenuse in to two segments of lengths 15cm and 20cm. The length of the arc of the semicircle between the points at which the legs touch the semicircle is $K\pi$. Then the numerical value of K is _____.
22. The value of $81^{\frac{1}{\log_3 3}} + 27^{\frac{1}{\log_9 36}} + 3^{\frac{1}{\log_3 9}}$ is _____.
23. The sum of all even two digits numbers is _____.
24. ABC is a triangle. D is the midpoint of AB. E is the midpoint of DB. F is the midpoint of BC. If the area of ΔABC is 64cm^2 , then the area of ΔAEF (in cm^2) is _____.
25. In a G.P of real numbers, the sum of the first two terms is 7. The sum of the first six terms is 91. The sum of the first four terms is _____.
26. In ΔADE , $\angle ADE = 140^\circ$. B and C are points on AD and AE respectively. A,B,C,D,E are all distinct. If $AB = BC = CD = DE$ then $\angle EAD$ is equal to _____.
27. When 265 is divided by a two digit number the remainder is 5. The number of such two digit numbers is _____.
28. Viswa is playing with three rectangular boxes which are identical. The dimensions of each are $50\text{ cm} \times 5\text{ cm} \times 2\text{ cm}$. He glues them in several ways. After gluing the maximum surface area he could get is _____ cm^2 .
29. A 5- digit integer is reversed. The difference of the original and the reversed number is not zero. The largest prime number that must be a factor of this difference is _____.
30. If $x^3 + 3xy^2 = 14$, $y^3 + 3yx^2 = 13$, x, y are real then the value of $x^2 + y^2$ is _____.
- _____()